R20

Code No.: 20CSC20

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. V Sem (Main) Examination Dec 2022 / Jan 2023

Operating Systems

(Common to CSE, CSE-AI&ML, CSE-IoT&CS Including BCT)

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A & Part –B (Internal Choice)** at one place in the same order

Part - A
(5Q X 3M = 15 Marks)

		M	CO	BT
1	Sketch the hierarchy of Storage device.	(3)	1	2
2	Provide the diagram showing context switch from process to process.	(3)	2	2
3	What is the purpose of page tables in paging?	(3)	3	2
4	What do you meant by Password Vulnerabilities?	(3)	4	2
5	What is Journaling?	(3)	6	1

Part – B
(5Q X 9M = 45 Marks)

			M	CO	BT
6	(a)	A computer system can be organized in to number of ways. Explain the different categories of computer systems that are categorized based on the number of general-purpose processors used.	(5)	1	2
	(b)	Compare the different categories of Operating-Systems based on their types of Operations.	(4)	1	2
		(OR)			
7	(a)	What is System call? Illustrate how system calls are used to provide operating system services.	(5)	1	2
	(b)	What is the main advantage of the layered approach to system design? What are the disadvantages of the layered approach?	(4)	1	2
8	(a)	Can a multithreaded solution using multiple user-level threads achieve better performance on a multiprocessor system than on a single-processor system? Explain.	(4)	2	3
	(b)	Consider the following set of processes, with the length of the CPU burst given in milliseconds:	(5)	2	3

Process ID	Burst Time	Priority
P_1	5	4
P_2	3	1
P_3	1	2
P_4	7	5
P_5	4	3

The processes are assumed to have arrived in the order P_1, P_2, P_3, P_4, P_5, all at time 0.

i. Draw Gantt chart that illustrate the execution of these processes using the non-preemptive priority scheduling algorithm (a larger priority number implies a higher priority).

ii. What is the turnaround time and waiting time of each process of the scheduling algorithm?

(OR)

9	(a)	What is the meaning of the term busy waiting? Can busy waiting be avoided altogether? Explain your answer.	(4)	2	3
	(b)	Explain the Banker's algorithm with illustrative steps for deadlock-avoidance.	(5)	2	2
10	(a)	Apply FCFS and SSTF disk scheduling algorithms on the following sequence of disk requests and calculate total head movement: 68,78,42,32,93,35,82. Assume initial position of head is at 40.	(5)	3	3
	(b)	Under what circumstances do page faults occur? Describe the actions taken by the operating system when a page fault occurs.	(4)	3	3

(OR)

11	(a)	Apply FIFO and Optimal Page Replacement algorithms on the following reference string 1,2,3,4,1,2,3,4,2,1,3,4 by considering 3 frames.	(5)	4	3
	(b)	Explain different methods for free space management.	(4)	4	2
12	(a)	Explain about the OS-level memory protection.	(5)	5	2
	(b)	Discuss a means by which managers of systems connected to the Internet could design their systems to limit or eliminate the damage done by worms. What are the drawbacks of making the change that you suggest?	(4)	5	3

(OR)

13	(a)	Buffer-overflow attacks can be avoided by adopting a better programming methodology or by using special hardware support. Discuss these solutions.	(5)	5	2
	(b)	Mobile operating systems such as iOS and Android place the user data and the system files into two separate partitions. Aside from security, what is an advantage of that separation?	(4)	5	2
14	(a)	Would you classify Linux threads as user-level threads or as kernel level threads? Support your answer with the appropriate arguments.	(5)	6	2
	(b)	Explain about process management in UNIX.	(4)	6	2

(OR)

15	(a)	Discuss about Passing of Data among Processes in Unix OS.	(5)	6	2
	(b)	Explain about IPC and network related system calls in UNIX.	(4)	6	2

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